Unique Common Neighbors Force a Universal Vertex

Proof Workbench

Statement version 0.2; reviewed in R001 and R002

Theorem 1. *Let G be a finite, nonempty, simple undirected graph such that every two distinct vertices have exactly one common neighbor. Then G has a vertex adjacent to every other vertex.*

Proof. If G has one vertex, the conclusion is vacuous. Suppose henceforth, for a contradiction, that G has no universal vertex.

For a vertex v, every $u \in N(v)$ has exactly one neighbor in the induced graph $G[N(v)]$: its unique common neighbor with v. Thus $G[N(v)]$ is 1-regular, so every degree of G is even. If u, v are nonadjacent, let w be their unique common neighbor. For each $x \in N(u) \setminus \{w\}$, map x to the unique common neighbor of x and v. This lies in $N(v)$ and is not u. The map is injective, for a common image of distinct x_1, x_2 would give the two distinct vertices u and that image as common neighbors of x_1, x_2. Hence $d(u) - 1 \leq d(v)$; symmetrically $d(v) - 1 \leq d(u)$. Since both degrees are even, $d(u) = d(v)$.

The complement $\overline{G}$ is connected. Otherwise, let X be one component and let $Y = V(G) \setminus X$. Every vertex of X is adjacent in G to every vertex of Y. Neither X nor Y has size one, since $\overline{G}$ has no isolated vertex. For $x \in X$ and $y \in Y$, the unique-common-neighbor condition gives

$$d_{G[X]}(x) + d_{G[Y]}(y) = 1.$$

Thus these internal degrees are constants a, b with $a + b = 1$. If $a = 0$, two vertices of X have all at least two vertices of Y as common neighbors; the case $b = 0$ is symmetric. Both contradict uniqueness. Along every edge of $\overline{G}$, the endpoints have equal degree in G, so complement connectivity makes G k-regular.

Let $n = |V(G)|$, and let A, J, I be respectively the adjacency, all-ones, and identity matrices of order n. Counting common neighbors gives

$$A^2 = J + (k-1)I.$$

Since $A\mathbf{1} = k\mathbf{1}$, applying this identity to $\mathbf{1}$ yields

$$k^2 = n + k - 1. \tag{1}$$

Every vertex has a neighbor, and k is even, so $k \geq 2$. Put $s = \sqrt{k-1}$ and let $W = \{z \in \mathbb{R}^n : \mathbf{1}^\mathsf{T} z = 0\}$. Because G is undirected, A is symmetric, and W is A-invariant. On W, $A^2 = s^2 I$. Hence each $z \in W$ is the sum of

$$z_+ = \tfrac{1}{2}(z + Az/s), \qquad z_- = \tfrac{1}{2}(z - Az/s),$$

which are respectively s- and $-s$-eigenvectors. If the corresponding eigenspace dimensions are p, q, then the zero diagonal of A gives

$$0 = \operatorname{tr}(A) = k + (p-q)s. \tag{2}$$

Here $p \neq q$, so s is rational. Since $s^2 = k - 1$ is an integer, s is a positive integer. Equation (2) gives $s \mid k$, while $k = s^2 + 1$, so $s \mid 1$. Therefore $s = 1$, $k = 2$, and (1) gives $n = 3$. The only simple 2-regular graph on three vertices is K_3, whose vertices are universal, contradicting the choice of G. $\square$

Review record. This source restates the proof recorded in `PROOF.md`. Fresh-context statement/logic review R001 and hypotheses/counterexample review R002 both passed; R001 prompted the explicit symmetry sentence used above.